

Skin Cancer Detection through Deep Learning using Skin Lesion Images

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Outline

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Introduction

- Skin cancer is among the most prevalent malignancies globally — incidence rising 33% from 2007–2017
- Melanoma projected to reach 510,000 new cases and 96,000 deaths by 2040 (Arnold et al., 2022)
- Manual diagnosis relies on visual inspection — subjective, time-consuming, specialist-dependent
- Global shortage of dermatologists causes significant delays in diagnosis and treatment
- Deep learning has demonstrated dermatologist-level performance in medical image classification
- **Project Goal: Automated 9-class skin lesion classification using EfficientNet-B2 with explainability**

Motivation

- Skin cancer remains one of the fastest-growing forms of cancer worldwide, emphasizing the need for early and accurate diagnosis.
- Increasing patient volume and limited availability of dermatology specialists often result in delayed diagnosis and treatment.
- Conventional diagnostic approaches rely heavily on expert interpretation and may suffer from inter-observer variability and limited sensitivity.
- Deep learning models enable automated, scalable, and consistent skin lesion classification from dermoscopic images.
- Integration of explainable AI methods such as Grad-CAM enhances interpretability and supports clinical confidence in AI-assisted diagnosis.

Literature Survey

S.No	Authors & Year	Model / Approach	Dataset	Limitation
1	Andre Esteva et al. (2017)	InceptionV3 + Transfer Learning	129K clinical skin images	Primarily binary classification; limited explainability (PMC)
2	Tobias Brinker et al. (2019)	ResNet50 Fine-Tuned CNN	ISIC 2016	Focused mainly on melanoma vs nevus classification
3	Mingxing Tan & Quoc Le (2019)	EfficientNet with Compound Scaling	ImageNet	General-purpose architecture; not medical-domain specific (arXiv)
4	Saket Chaturvedi et al. (2019)	MobileNet Transfer Learning	HAM10000	Moderate performance on multiclass classification (arXiv)
5	Yeong Chan Lee et al. (2018)	Fine-Tuned Neural Network Ensemble	ISIC Challenge 2018	Complex ensemble architecture increases computation (arXiv)
6	V Venugopal et al. (2023)	Modified EfficientNet + DNN	Combined dermoscopic datasets	Binary classification emphasis; higher training complexity (ScienceDirect)
7	A Naeem et al. (2024)	CNN + Handcrafted Feature Fusion	ISIC 2019	Increased feature engineering complexity (MDPI)
8	K Kanchana et al. (2024)	EfficientNet B0–B7 Transfer Learning	EfficientNet skin lesion dataset	Performance sensitive to preprocessing pipeline (PubMed)
9	Green Arther Sandag et al. (2025)	EfficientNet + Grad-CAM	Kaggle dermoscopic dataset	Limited to binary melanoma classification (ResearchGate)
10	S Islam et al. (2026)	EfficientNet-B2 + Metadata Fusion	79,246 UK patient images	Proprietary dataset; limited public reproducibility (Nature)

Research Gaps

Below are the Identified Gaps from Literature and How We Addressed Them

Research Gap	Our Contribution
Class Imbalance	Used WeightedRandomSampler + Class-weighted CrossEntropyLoss
Lack of Explainability	Grad-CAM heatmaps on final convolutional layer
Overfitting on Small Data	Transfer learning (ImageNet) + Augmentation + AdamW weight decay
Binary-Only Classification	9-class simultaneous multi-class classification

Objectives

1. To develop a robust multi-class skin lesion classification system capable of identifying nine major categories of skin diseases using the ISIC dataset.
2. To implement EfficientNet-B2 with ImageNet pretrained weights through transfer learning for efficient and accurate feature extraction.
3. To design an effective image preprocessing and data augmentation pipeline for improving model generalization on limited medical image datasets.
4. To address class imbalance using WeightedRandomSampler and class-weighted loss functions to ensure balanced learning across all lesion categories.
5. To comprehensively evaluate model performance using metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC analysis.
6. To integrate Grad-CAM-based explainability for generating clinically interpretable visualizations that support trustworthy AI-assisted diagnosis.

Dataset

Class	Train	Test
Actinic Keratosis	114	16
Basal Cell Carcinoma	376	16
Dermatofibroma	95	16
Melanoma	438	16
Nevus	357	16
Pigmented Benign Keratosis	462	16
Seborrheic Keratosis	77	3
Squamous Cell Carcinoma	181	16
Vascular Lesion	139	3

9

Classes

2357

Total Images

2239

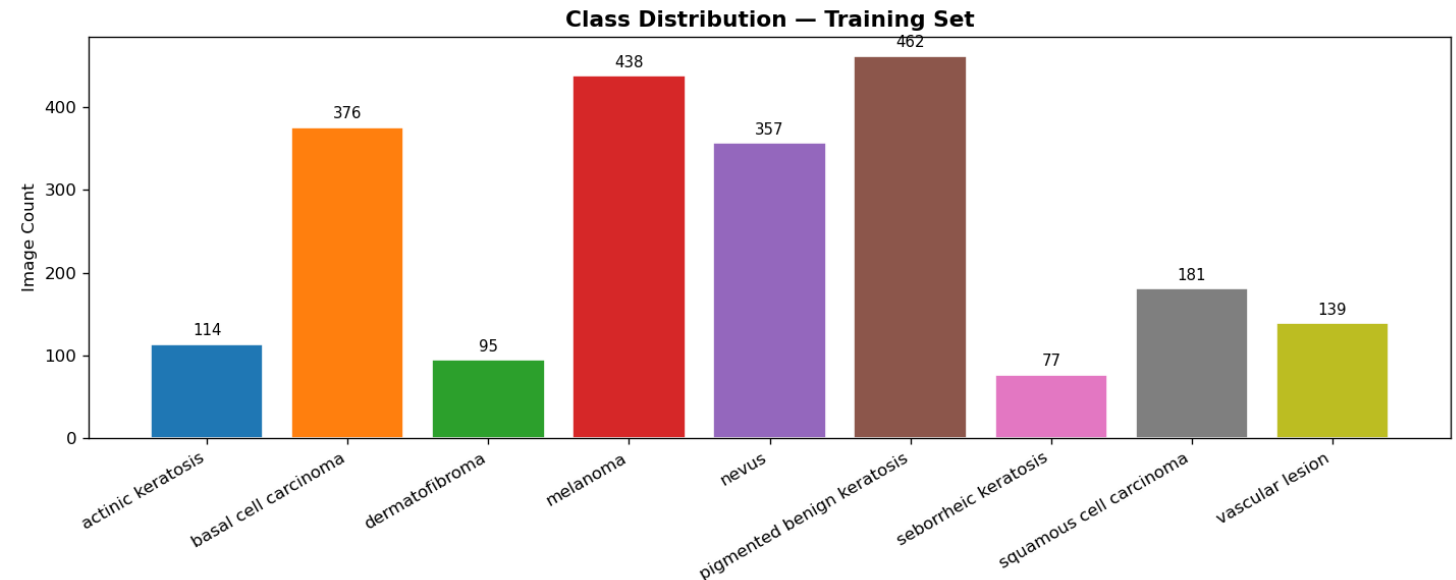
Train Images

118

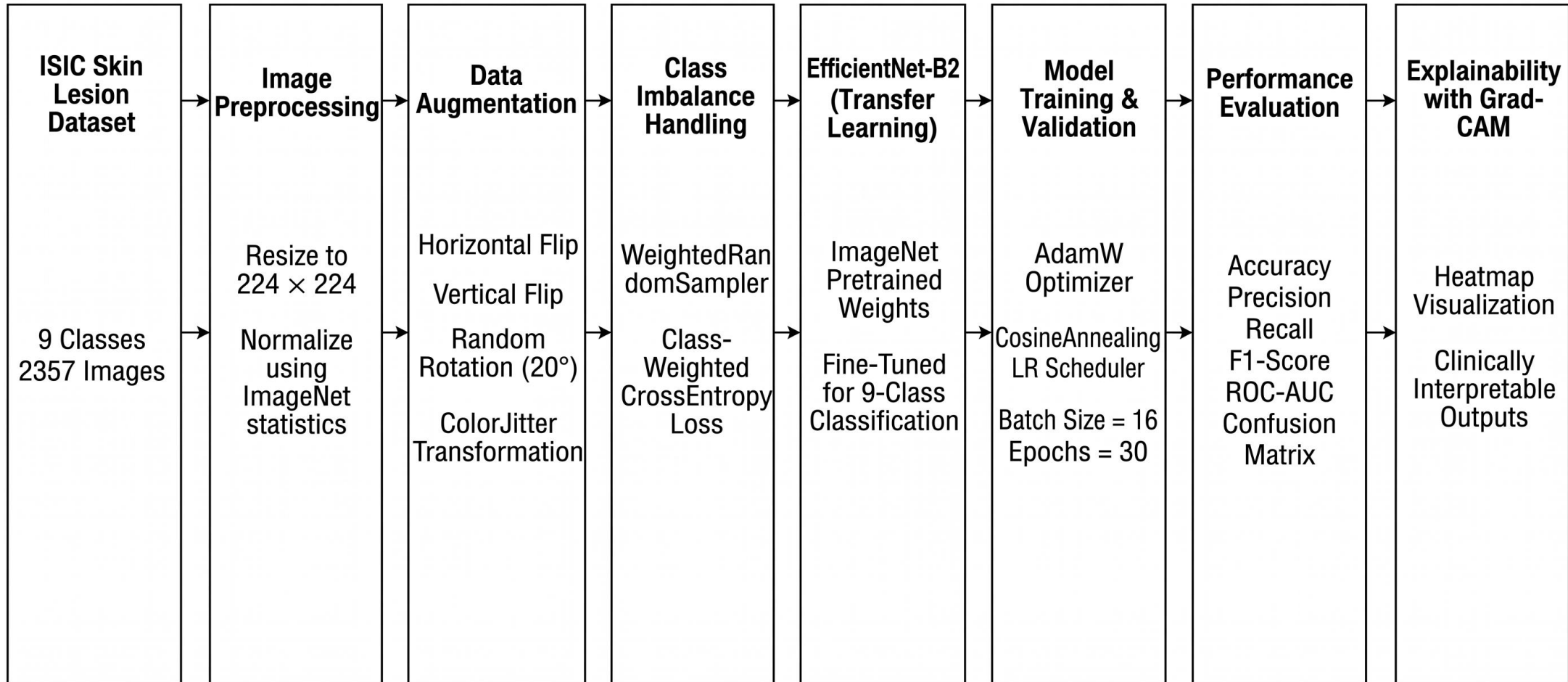
Test Images

85/15

Train/Val Split



Flowchart



Baseline CNN Architecture

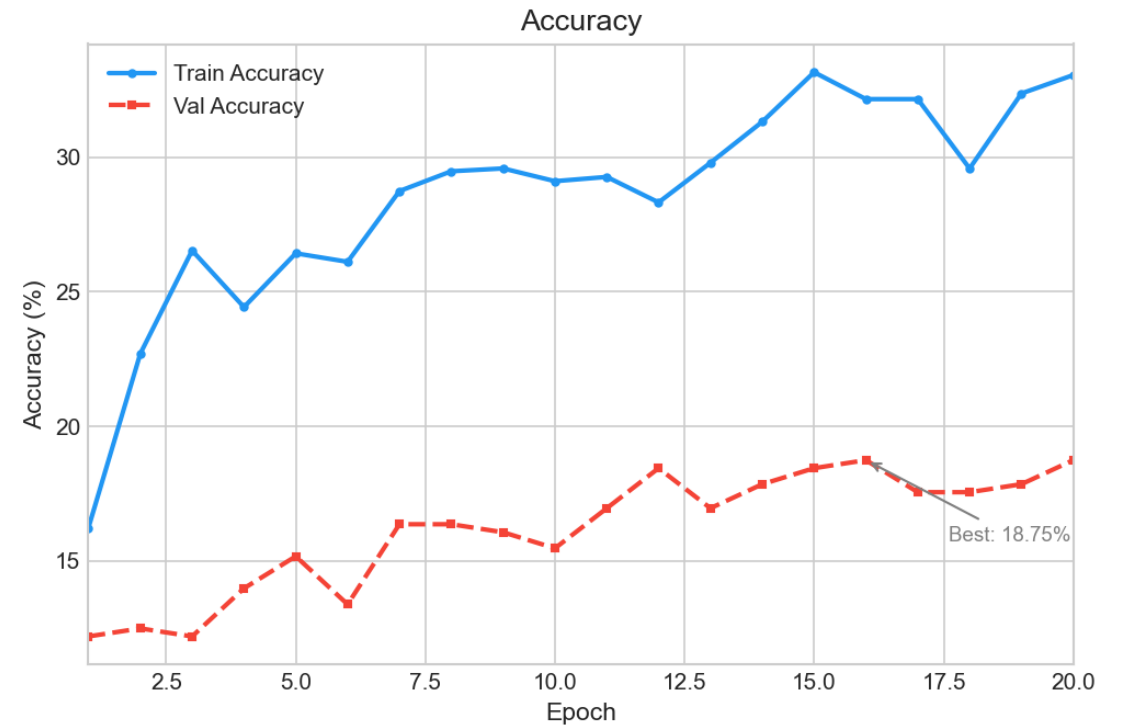
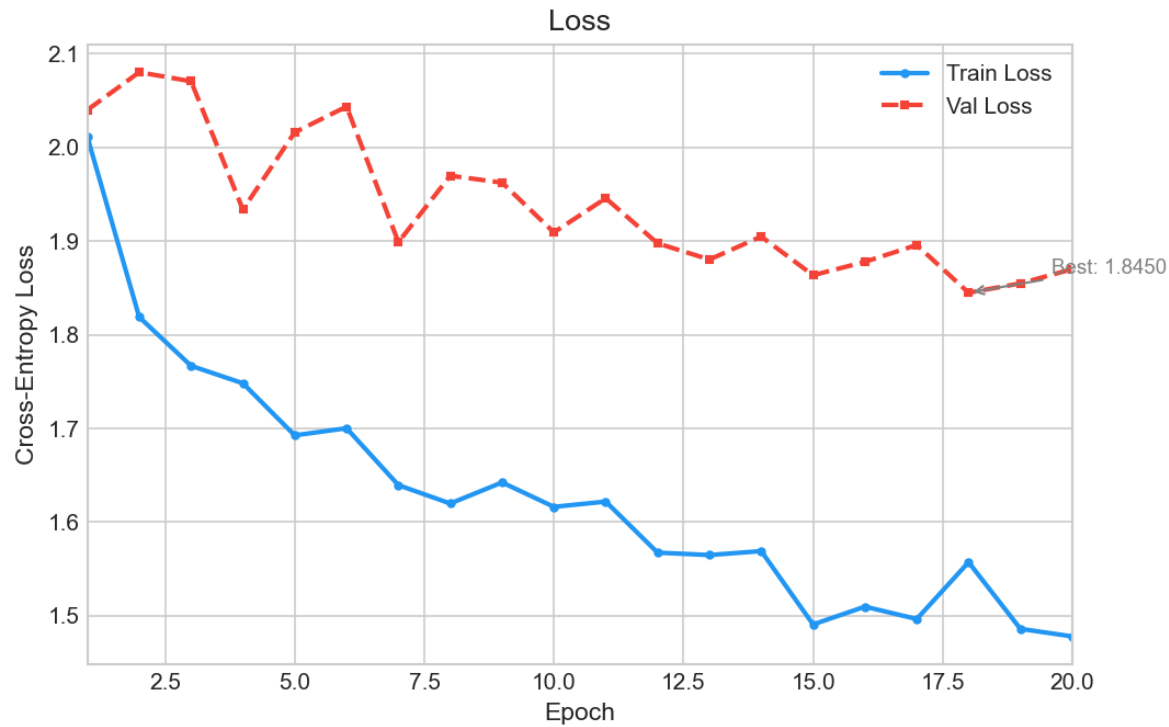
Baseline CNN Flowchart

Input → Conv → ReLU → MaxPool → Conv → ReLU → MaxPool → FC → Output

- A custom Baseline CNN model was developed for performance comparison.
- The architecture consists of multiple convolutional layers followed by ReLU activation and MaxPooling operations.
- Dropout layers were used to reduce overfitting and improve generalization.
- Fully connected dense layers were used for final 9-class classification.
- The Baseline CNN provides a simpler reference model against which EfficientNet-B2 performance can be evaluated.

Baseline CNN Training Curve

Training Curves — BaselineCNN



EfficientNet-B2 Architecture

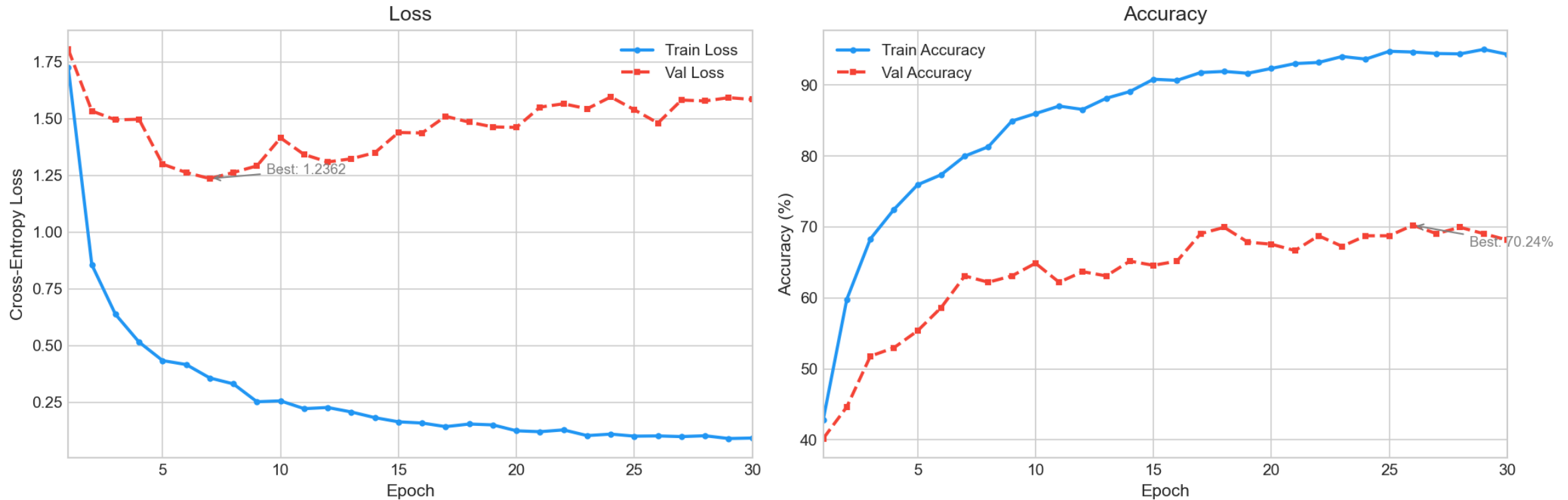
What is EfficientNet?

EfficientNet is a Convolutional Neural Network (CNN) architecture developed by Google for efficient and accurate image classification. It uses a technique called compound scaling, which uniformly scales the network's depth, width and image resolution

- EfficientNet-B2 is a CNN architecture optimized for efficient image classification.
- Uses compound scaling to balance network depth, width, and resolution.
- Built using MBConv blocks with Squeeze-and-Excitation (SE) attention.
- Transfer learning applied using ImageNet pretrained weights.
- Final classification layer modified for 9-class skin lesion classification.
- Provides high accuracy with lower computational complexity.

EfficientNet-B2 Training Curve

Training Curves — EfficientNet-B2



Grad-CAM Explainability

What is Grad-CAM?

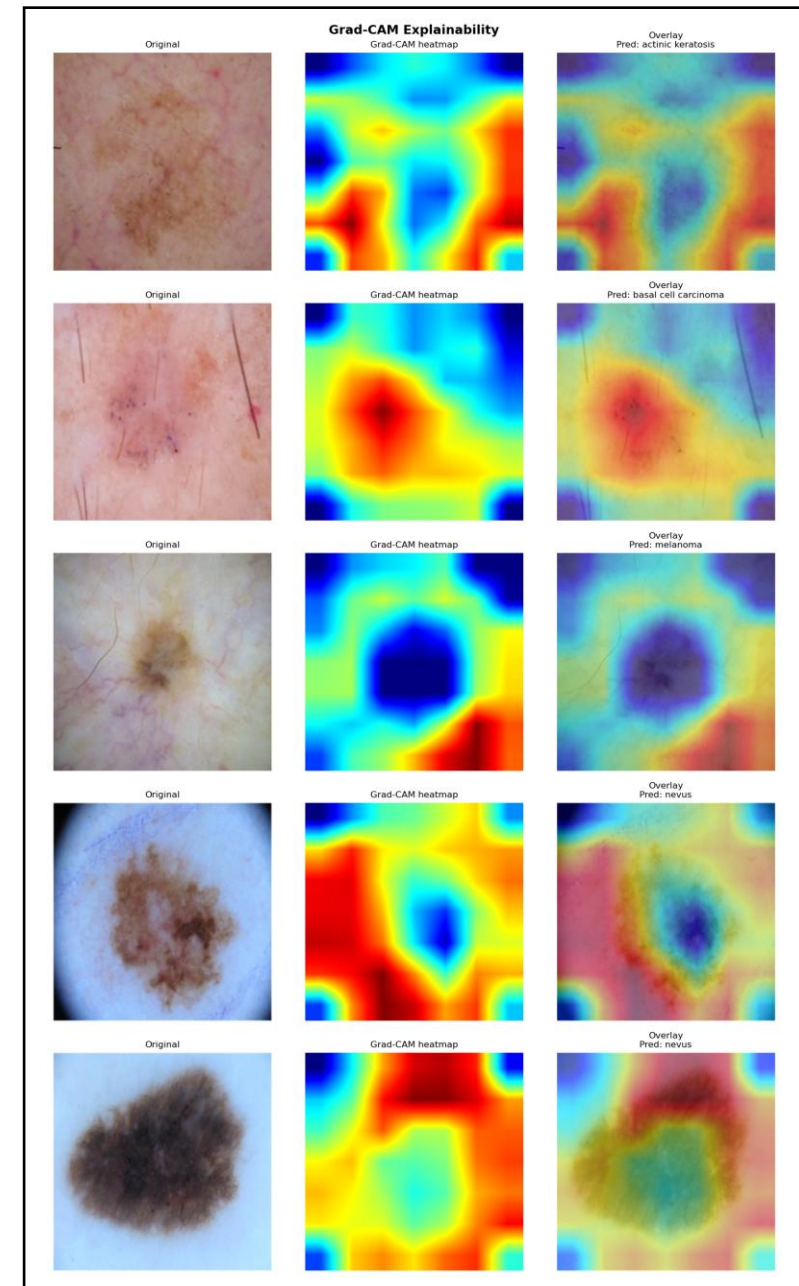
It is an explainable AI technique used to visualize model decision-making. It highlights important image regions that contribute most to the predicted class. Helps interpret how the CNN focuses on skin lesion features during classification.

What Grad-CAM Shows

Red/yellow regions indicate areas receiving high model attention. Blue/dark regions indicate less important areas.

How It Works

Feature maps are weighted based on their contribution to the prediction. A heatmap is generated and overlaid on the original image.



Test Results – Classification Report

BaselineCNN – Test Results

Test Loss: 2.1219
Test Accuracy: 29.66%

Classification Report:

	precision	recall	f1-score	support
actinic keratosis	0.4000	0.6250	0.4878	16
basal cell carcinoma	0.0000	0.0000	0.0000	16
dermatofibroma	0.1739	0.5000	0.2581	16
melanoma	0.0000	0.0000	0.0000	16
nevus	0.5500	0.6875	0.6111	16
pigmented benign keratosis	0.0000	0.0000	0.0000	16
seborrheic keratosis	0.1765	1.0000	0.3000	3
squamous cell carcinoma	0.0000	0.0000	0.0000	16
vascular lesion	0.6000	1.0000	0.7500	3
accuracy			0.2966	118
macro avg	0.2112	0.4236	0.2674	118
weighted avg	0.1721	0.2966	0.2107	118

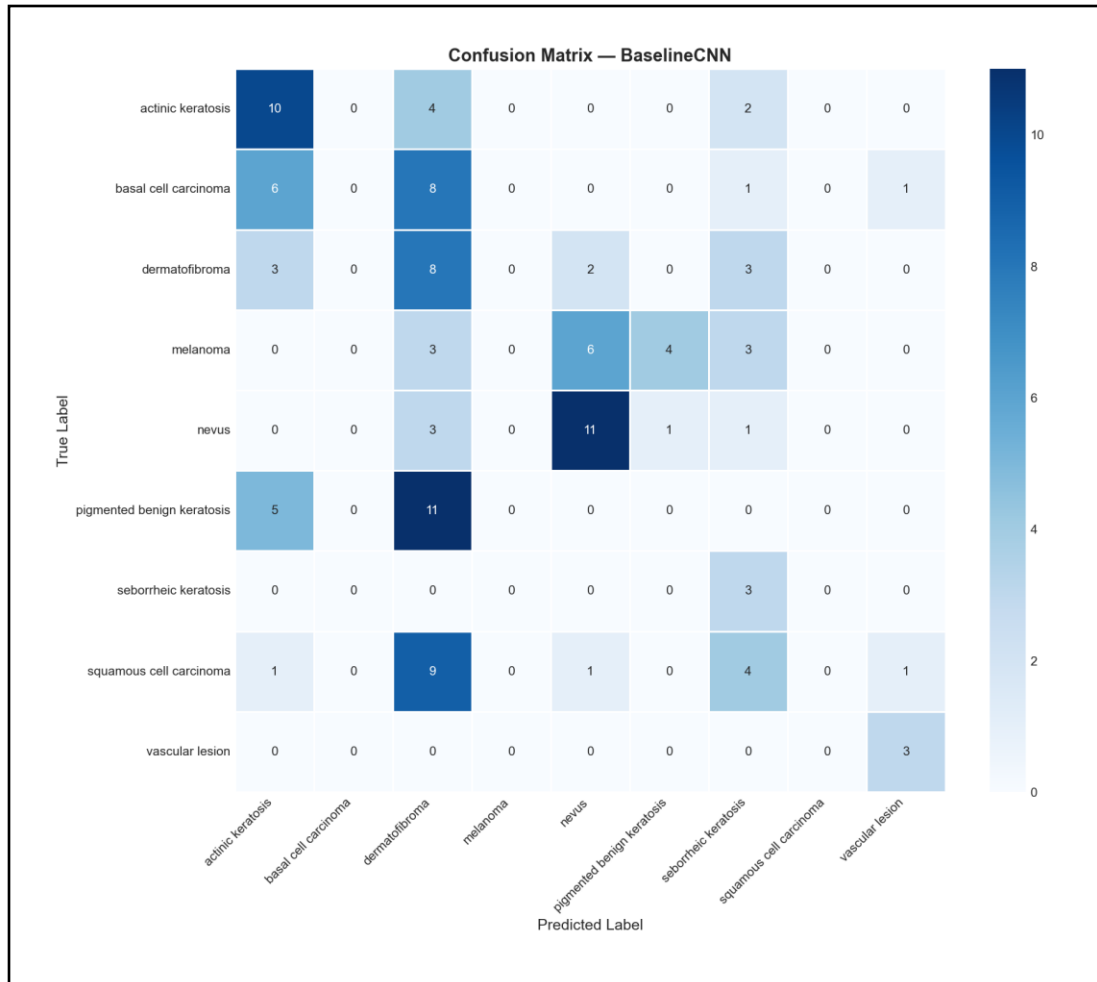
EfficientNet-B2 – Test Results

Test Loss: 2.3283
Test Accuracy: 59.32%

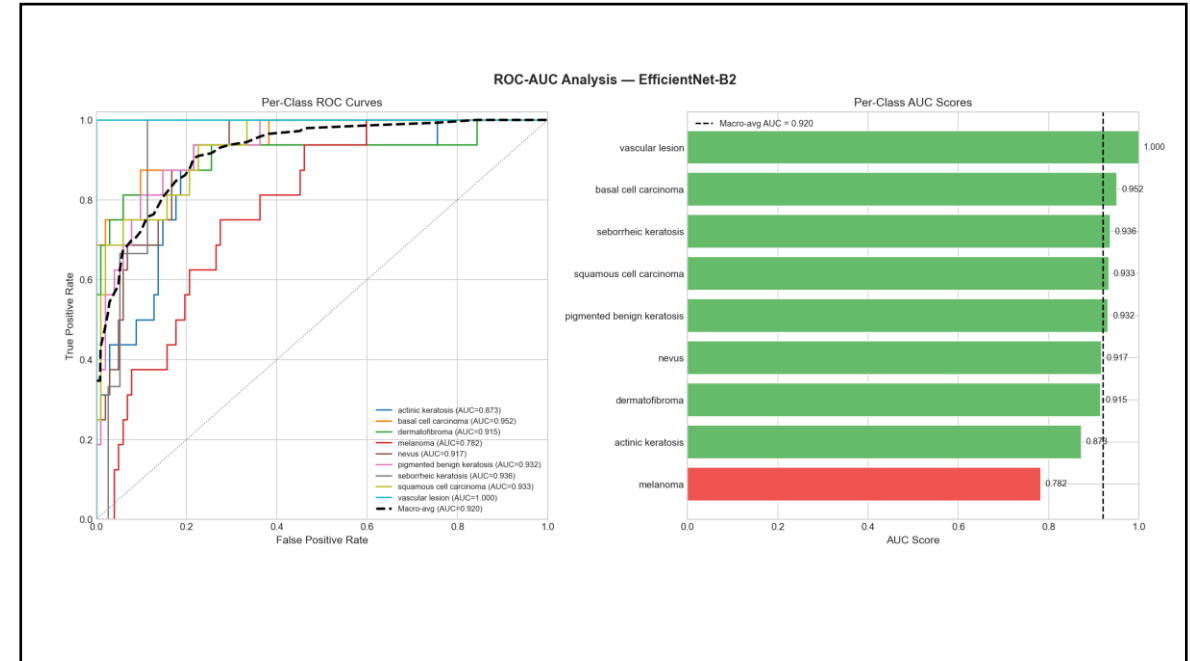
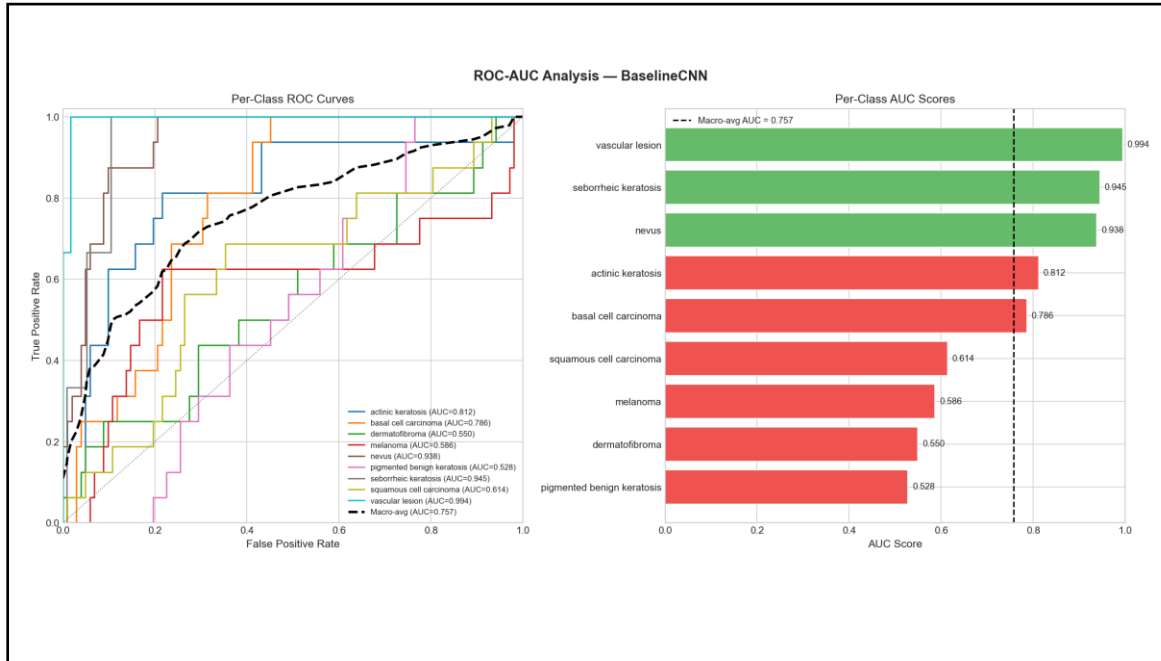
Classification Report:

	precision	recall	f1-score	support
actinic keratosis	0.8333	0.3125	0.4545	16
basal cell carcinoma	0.6500	0.8125	0.7222	16
dermatofibroma	0.9167	0.6875	0.7857	16
melanoma	0.3333	0.1875	0.2400	16
nevus	0.3659	0.9375	0.5263	16
pigmented benign keratosis	0.6667	0.6250	0.6452	16
seborrheic keratosis	0.0000	0.0000	0.0000	3
squamous cell carcinoma	0.8333	0.6250	0.7143	16
vascular lesion	1.0000	1.0000	1.0000	3
accuracy			0.5932	118
macro avg	0.6221	0.5764	0.5654	118
weighted avg	0.6490	0.5932	0.5798	118

Test Results – Confusion Matrix



Test Results – ROC AUC



Conclusion

01 — Effective Multi-Class Classification

Successfully developed a deep learning–based 9-class skin lesion classification system using EfficientNet-B2 and transfer learning on the ISIC dataset.

02 — Improved Feature Learning through Transfer Learning

ImageNet pretrained weights enabled efficient feature extraction and improved classification performance on limited medical imaging data.

03 — Robust Handling of Class Imbalance

WeightedRandomSampler and class-weighted loss functions improved learning across minority lesion categories and reduced prediction bias.

04 — Strong Model Evaluation and Interpretability

Comprehensive evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix demonstrated reliable model performance, while Grad-CAM provided clinically interpretable visual explanations.

Future Work

Metadata Integration

Incorporate patient clinical information such as age, gender, lesion location, and medical history to improve diagnostic accuracy and contextual understanding.

Ensemble Learning Approaches

Combine multiple deep learning architectures to improve robustness, generalization, and overall classification performance.

Clinical Validation and Real-World Testing

Evaluate the model on larger real-world clinical datasets and compare performance with dermatologist assessments for practical healthcare adoption.

References

- [1] S. Islam, G. C. Wishart, J. Walls, P. Hall, A. G. Seco de Herrera, J. Q. Gan, and H. Raza, “*Advancing Skin Cancer Detection Through Deep Learning and Fusion of Patient Metadata and Skin Lesion Images*,” Scientific Reports, vol. 16, no. 1968, 2026, doi: 10.1038/s41598-025-26392-4.
- [2] A. Esteva et al., “*Dermatologist-level classification of skin cancer with deep neural networks*,” Nature, vol. 542, no. 7639, pp. 115–118, 2017.
- [3] T. Brinker et al., “*Comparing artificial intelligence algorithms to 157 German dermatologists: the melanoma classification benchmark*,” European Journal of Cancer, vol. 111, pp. 30–37, 2019.
- [4] M. Tan and Q. Le, “*EfficientNet: Rethinking model scaling for convolutional neural networks*,” in Proc. ICML, 2019, pp. 6105–6114.
- [5] J. Selvaraju et al., “*Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*,” in Proc. ICCV, 2017, pp. 618–626.
- [6] N. Codella et al., “*Skin Lesion Analysis Toward Melanoma Detection: A Challenge at the International Symposium on Biomedical Imaging*,” ISBI, 2018.
- [7] H. A. Haenssle et al., “*Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition*,” Annals of Oncology, vol. 29, no. 8, pp. 1836–1842, 2018.

References

- [8] P. Tschandl et al., “*Human–computer collaboration for skin cancer recognition*,” *Nature Medicine*, vol. 26, pp. 1229–1234, 2020.
- [9] A. Dosovitskiy et al., “*An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*,” *ICLR*, 2021.
- [10] ISIC Archive, “*International Skin Imaging Collaboration (ISIC) Dataset*,” [Online]. Available: <https://www.isic-archive.com/>
- [11] K. He et al., “*Deep Residual Learning for Image Recognition*,” in *Proc. CVPR*, 2016, pp. 770–778.
- [12] O. Russakovsky et al., “*ImageNet Large Scale Visual Recognition Challenge*,” *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015.

Thank You